EE 102: Signal Processing and Linear Systems

Problem Set #5: Applying Convolution

Name: _____ **Submission Date:** _____

Problem 1 A system responds to an impulse input $\delta(t)$ in an exponentially decaying manner. So, the impulse response of the system is given by:

$$h(t) = e^{-2t}u(t)$$

where $u(t)$ is the unit step function.

(a) [10 points] Is the system linear and time-invariant? Justify your answer.

(b) [10 points] What is the output $y(t)$ of the system when the input is $x(t) = k\delta(t)$, where k is a constant?

(c) [20 points] What is the output $y(t)$ of the system when the input is $x(t) = u(t)$? You must find this by starting from the relationship between the step signal and the impulse signal (from problem set #2):

$$u(t) = \int_{-\infty}^{t} \delta(\tau)d\tau$$

Then, use linearity and the impulse response $h(t)$ to find the output $y(t)$.

(d) [20 points] Find the output $y(t)$ of the system when the input is a pulse signal of amplitude A and duration τ, $\quad x(t) = A[u(t) - u(t-\tau)]$.

(e) [10 points each] Find the output of the system to the following input signals

1. $x(t) = e^{-3t}u(t)$
2. $x(t) = \cos(2t)u(t)$
3. $x(t) = \sin(4t)u(t)$
4. $x(t) = tu(t)$